

Moire Hurwitz Diophantine Search

Mega Huge Deconstruction:

Moire–cusp search, arithmetic encoding, and Diophantine bookkeeping for structured witnesses

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Abstract

This note maps the **formally proved** fragments of the HQIV Lean library that touch a **geometric SAT demo** pipeline (`scripts/hqiv_geometric_3sat_demo.py`). The **only** supported **patch search geometry** in the Lean corpus is **one-dimensional** in the patch index j : **moiré cumulative variation**, **cusp / threshold crossing**, and **binary search** (`Hqiv/Algebra/MoireCuspBracket.lean`, `Hqiv/Algebra/MoireToyThresholdSearch.lean`). Fourier correlations, Ω -axis angles, and integer-shell embeddings feed the **same** ordered-index score; they are **not** an auxiliary “quantum ray” or 2D vector search (the old `OctonionPatchQuantum` track has been **removed** from the library). The acronym **MHD** is **Mega Huge Deconstruction**: an explicit inventory of proved logic, analytic hooks, and open encoding bridges—without conflating BST on a monotone trace with worst-case SAT over 2^n assignments.

Repository. <https://github.com/disregardfiat/hqiv-lean>

Reader guide. **Green (proved):** Lean identifiers cited inline. **Amber (scaffolds):** Fourier concentration bundles, encoding completeness for abstract P . **Red (gap):** no theorem identifies moiré diagnostics with `CNFFormula.eval` in general.

1 Introduction: patch search = moiré–cusp

Classical NP witnesses live in a huge discrete space. The HQIV pipeline separates:

1. **Semantic SAT** (truth of a CNF).
2. **Ordered patch search** along $j \in \{0, \dots, n_{\text{patch}} - 1\}$ using **monotone** cumulative variation and BST — the **moiré–cusp** layer.

No claim: $P = NP$. **Claim in Lean:** the discrete search mechanics for the patch are **order-theoretic** (threshold crossings, least indices). **Retired narrative:** orthogonal “quantum rays” or 2D slice search as an alternative patch geometry; the library no longer contains `OctonionPatchQuantum`.

“Hurwitz” and “Diophantine”. Division-algebra / Hurwitz language may appear in other HQIV scaffolds (`Hqiv/Physics/DivisionAlgebraZetaScaffold.lean`). **Diophantine** here means Ω -style literal weights and shell integers (`Hqiv/Logic/CNF.lean`), not Matiyasevich.

2 Logic and assignment indexing (combinatorial layer)

Hqiv/Logic/CNF.lean: literals, clauses, evaluation, satisfiability, ω -encoding weights.

Hqiv/Logic/CNFPatchBridge.lean: assignmentFromPatchIndex;

CNFFormula.satisfiable_iff_exists_patch_index reindexes assignments by $j \in \text{Fin}(2^n)$ (Python order). **This is reindexing, not a poly-time finder.**

Hqiv/Logic/GeometricEncodingCompleteness.lean: abstract null-pattern $P(\phi, k)$; encodingCompleteness_iff_null_globally_implies_unsat; corollaries for “all k null $\Rightarrow$ UNSAT” under an encoding-completeness hypothesis.

3 Moiré-cusp: the formal patch search layer

Hqiv/Algebra/MoireCuspBracket.lean: moireCumulativeAbsVariation (monotone in j), split lemmas, triangle / Gaussian-lens step on second differences.

Hqiv/Algebra/MoireToyThresholdSearch.lean: upward closure of “cum[j] $\geq T$ ”, existence of a least index above threshold — the **BST** interface. **Explicit:** no link to SAT in this file.

Hqiv/Algebra/MoireJerkSphereModeBridge.lean: discrete jerk of a patch score; closed forms for sinusoid models; ties Ω to jerk factors when the score is the intrinsic-axis sinusoid. Module doc now states that **patch search** is the moiré-cusp story, not spatial vector search.

Conclusion (proved). Along the patch, the **search** object is **one-dimensional** and **order-theoretic** in j ; outer k -sweeps are indexed diagnostics (null vs non-null), not a replacement for semantic SAT.

4 Analytic hooks: Ω , $\pi/(2k)$, Fourier sums (supporting the 1D score)

Hqiv/Algebra/OctonionAxisAngles.lean: axisAngle, intrinsicShellAxisAngle, two_mul_axisAngle_eq_pi_div_k.

Hqiv/Algebra/OctonionSphereFourierPatch.lean: peak/side correlations, KernelIsHarmonic, FourierPatchConcentration, discrete roots-of-unity sums. **Character orthogonality** here means Fourier / DFT orthogonality on $\mathbb{Z}/n\mathbb{Z}$ — not the removed ray geometry.

These supply **phase and spectral** structure for the **same** real score sampled along j . They do **not** assert a second independent “search direction” in $\mathbb{R}^8$ for the patch pipeline.

5 Synthesis and gaps

Encoding completeness for a concrete P instantiated from moiré diagnostics remains a **paper / analysis target**. **Target 0** (toy vs. Lean Fourier score) remains open; see AGENTS/FT_PATCH_CLOSED_TARGET.md.

Topic	Status
Moiré-cusp / BST on j	Proved order theory (MoireCuspBracket, MoireToyThresholdSearch).
SAT $\leftrightarrow$ score	Not proved.
Quantum ray / 2D slice search	Removed from Lean (OctonionPatchQuantum deleted).

Acknowledgement. Rebuild: lake build HQIVLEAN.